

Name:

Roll No:

INSTITUTE OF BUSINESS ADMINISTRATION  
 Department of Computer Science, Karachi  
 Course: Introduction to Artificial Intelligence (CSE-307)  
 Instructor: Dr. Syed Ali Raza, Sumaira Saeed, and Tahir Syed  
 Final Exam – Spring 2025

Duration: 3 hours

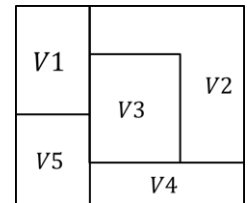
Total marks: 100

Instructions: Provide answers in the space provided after each question. You may use an extra sheet for rough work, but not for providing answers. Please answer using a pen only (pencils are not permitted).

### CONSTRAINT SATISFACTION PROBLEM

- 1) [10 marks] Color the map with 5 variables, V1, V2, V3, V4, V5, using the colors Red, Green, Blue, and Yellow. Write below the first solution found by the Backtracking algorithm. Write the color assigned to each variable.

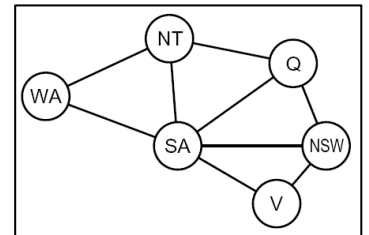
- Domain: {Red, Green, Blue, Yellow}
- Constraints: Adjacent variables must have different colors
- Variable order: V3, V5, V1, V2, V4
- Color order: Yellow, Green, Blue, Red



Answer:

| V3 | V5 | V1 | V2 | V4 |
|----|----|----|----|----|
|    |    |    |    |    |

- 2) [8 marks] For the map coloring problem of Australia using three colors (Red, Green, and Blue), you are given the starting domains of the six variables (the connected states of Australia shown in the constraint graph here) in the table below. A dash (-) in the domain values mean that the color is unavailable for that variable. For example, R is unavailable for NT, NSW, and SA. Your task is to apply the forward checking method in filtering and identify for the following two scenarios which variable's domain shrinks to an empty set first (i.e., the backtracking is needed) and after which variable's assignment (i.e., which variable's assignment has caused the other variable's domain to become empty). The order of picking colors for a variable's assignment is also given for each scenario.



| variables        | WA |   |   | NT |   |   | Q |   |   | NSW |   |   | V |   |   | SA |   |   |
|------------------|----|---|---|----|---|---|---|---|---|-----|---|---|---|---|---|----|---|---|
| Starting Domains | R  | G | B | -  | G | B | R | G | B | -   | G | B | R | G | B | -  | G | B |

| Scenario no. | Variable Orderings    | Color assignment ordering | Variable with the empty domain | Variable whose assignment caused the empty domain |
|--------------|-----------------------|---------------------------|--------------------------------|---------------------------------------------------|
| 1            | Q, NSW, SA, WA, V, NT | G, B, R                   |                                |                                                   |
| 2            | WA, V, NT, Q, NSW, SA | B, G, R                   |                                |                                                   |

- 3) [6 marks] Consider the map coloring problem of Australia using three colors (Red, Green, and Blue), as discussed in the previous question. You are given the starting domains of the six variables (shown in the constraint graph above) in the table below. A dash (-) in the domain values mean that the color is unavailable for that variable. For example, R and B are unavailable for NT at the start. For the following sequence of arcs, your task is to identify for which arc making it consistent causes the domain of another variable reduce to an empty set. Also, which variable's domain gets empty first?

| Arc              | WA |   |   | NT |   |   | Q |   |   | NSW |   |   | V |   |   | SA |   |   |
|------------------|----|---|---|----|---|---|---|---|---|-----|---|---|---|---|---|----|---|---|
| Starting domains | R  | G | - | -  | G | - | R | G | - | -   | G | B | R | G | B | R  | - | - |

Arcs to check in this sequence:

| Arc1    | Arc2   | Arc3    | Arc4   | Arc5    | Arc6    | Arc7   | Arc8   | Arc9    | Arc10   |
|---------|--------|---------|--------|---------|---------|--------|--------|---------|---------|
| SA → WA | V → SA | WA → NT | SA → Q | SA → NT | WA → SA | Q → NT | Q → SA | WA → NT | WA → NT |

Answers:

Write arc which makes a variable's domain empty:

Write variable whose domain becomes empty:

#### UNSUPERVISED LEARNING

- 4) [10 marks] For the data given here (9 instances of X and Y coordinates), apply the K-means algorithm for one iteration using K=3. Note: one iteration means finding the new centroids once. Use instance 3, 6, and 8 as the initial centroids. Use Manhattan distance as the distance measure.

$$\text{Manhattan Distance} = |x_1 - x_2| + |y_1 - y_2|$$

- a. Write below the updated centroids after the first iteration of K-means.

| Updated centroids | X | Y |
|-------------------|---|---|
| C1                |   |   |
| C2                |   |   |
| C3                |   |   |

| Instance | X  | Y  |
|----------|----|----|
| 1        | 4  | 7  |
| 2        | -1 | 3  |
| 3        | 2  | -2 |
| 4        | 5  | 1  |
| 5        | 7  | 6  |
| 6        | -5 | -2 |
| 7        | 3  | 6  |
| 8        | -3 | -1 |
| 9        | 6  | 0  |

- b. Calculate the Within cluster sum of squares (WSS) for K=3 (using the clusters and the centroids found above) using the Euclidian distance (not the Manhattan distance). Note, WSS is also known as the sum of squared error (SSE).

Answer: \_\_\_\_\_

- c. If you are given that the WSS for K=1 (all points in one cluster) is 242, then mention which K value is better and why?

Answer: \_\_\_\_\_

- 5) [2 marks] Explain why the notion of proximity (closeness or distance) between examples plays a central role in the K-means algorithm.

Answer:

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- 6) [2 marks] Which of the following best explains why K-means clustering tends to form hyperspherical clusters? Note, a hypersphere is a sphere in higher dimensions (4D or higher). Circle the correct option.
- A. It uses cosine similarity, which groups points based on angles rather than distance.
- B. It minimizes Manhattan distance between data points and cluster centers.
- C. It uses Euclidean distance and assumes that clusters are distributed symmetrically around a centroid.
- D. It explicitly models non-convex and elliptical shapes using covariance matrices.

### SUPERVISED LEARNING

- 7) [10 marks] The table below shows fictional statistics of student grades (pass and fail) in an AI course. There are eight examples with three characteristics found important in determining the success of a student, namely Game Skills, Favorite Campus and Favorite Drink. Assume we induce a Decision Tree (partially) using the Information Gain as the measure of impurity. Hint: Information gain = (information before split) – (information after split)

$$Entropy(t) = - \sum_j p(j|t) \log p(j|t)$$

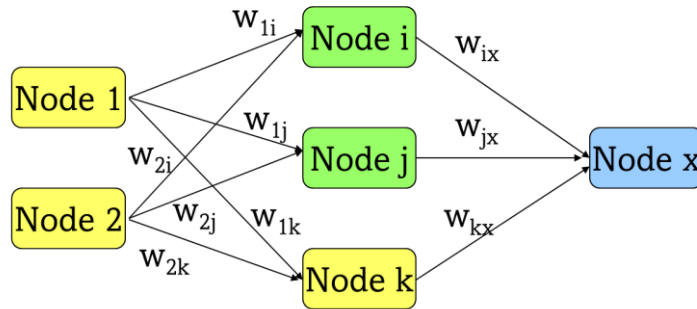
| Example | Game Skills | Favorite Campus | Favorite Drink | Grade       |
|---------|-------------|-----------------|----------------|-------------|
| X1      | Expert      | Main            | Next           | <b>Fail</b> |
| X2      | Novice      | Main            | Fizup          | <b>Fail</b> |
| X3      | Novice      | City            | Next           | <b>Pass</b> |
| X4      | Novice      | City            | Fizup          | <b>Fail</b> |
| X5      | Expert      | City            | Fizup          | <b>Fail</b> |
| X6      | Novice      | City            | Pakola         | <b>Pass</b> |
| X7      | Novice      | Main            | Next           | <b>Fail</b> |
| X8      | Expert      | City            | Next           | <b>Pass</b> |

Answer the following questions.

|                                                                                                  |  |
|--------------------------------------------------------------------------------------------------|--|
| What is the Information Gain for the attribute Game Skills?                                      |  |
| What is the Information Gain for the attribute Favorite Campus?                                  |  |
| What is the Information Gain for the attribute Favorite Drink?                                   |  |
| Which attribute is selected at the root?                                                         |  |
| Which branch or branches of the root node has a decision (Pass or Fail) on it (i.e., leaf node)? |  |

- 8) [8 marks] Consider the fully connected feed-forward neural network given below. Here, a node represents a single neuron.
- The initial wright values are  $w_{1i} = 0.20$ ,  $w_{1j} = 0.10$ ,  $w_{1k} = 0.30$ ,  $w_{2i} = -0.10$ ,  $w_{2j} = -0.10$ ,  $w_{2k} = 0.20$ ,  $w_{ix} = 0.10$ ,  $w_{jx} = 0.50$ ,  $w_{kx} = 0.20$ .

- Input is node1 = 0.4 and node2 = 0.6, and the target value is  $T=0.65$ . The learning rate ( $r$ ) is 0.5. For all the calculations, use values till 5 decimal places.
- Use the sigmoid function as the activation,  $f(x) = 1 / (1 + e^{-x})$ .
- The formula for the error at node  $x$  is  $\text{Error}(x) = (T - O_x) O_x (1 - O_x)$  where  $O_x$  is the output of node  $x$ .
- The formula to compute new weight between two nodes  $a$  and  $b$  is,
  - $w_{ab} = w_{ab} (\text{current}) + \Delta w_{ab}$
  - where,  $\Delta w_{ab} = r \times \text{Error}(b) * O_a$



Answer the following questions:

|                                                |  |
|------------------------------------------------|--|
| What is the output at node $x$ ?               |  |
| What is the error at node $x$ ?                |  |
| What is the updated value of weight $w_{kx}$ ? |  |

- 9) [2 marks] We train models using a dataset, but we ultimately want them to work well on data they haven't seen. What do we call this goal, and why is it central to the learning process? Answer briefly.

Answer:

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- 10) [2 marks] When a decision tree algorithm has two possible ways to split the data, how does it decide which split to choose? Circle the right option.

- A. It chooses the split that maximizes the number of branches.
- B. It chooses the split that creates the smallest number of examples in each branch.
- C. It chooses the split that results in the most homogeneous (pure) branches.
- D. It chooses the split that minimizes the total number of features used.

- 11) [2 marks] What is the main goal of cross-validation?

Answer:

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- 12) [6 marks] Consider the following confusion matrix of a predictive model to detect spam e-mail. Find the values of the following measures:

|                |          | PREDICTION |          |
|----------------|----------|------------|----------|
|                |          | Spam       | Not Spam |
| ACTUAL         | Spam     | 560        | 40       |
|                | Not Spam | 50         | 50       |
| Precision      |          |            |          |
| Recall         |          |            |          |
| F-measure (F1) |          |            |          |

#### KNOWLEDGE REPRESENTATION (LOGICAL REASONING)

- 13) [2 marks] What is the main limitation of the propositional logic?

Answer:

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- 14) [8 marks] Assume that several facts of the following types are already defined using First Order Logic:

male(X)

female(X)

parent(X, Y)

For example, male(albert). female(elizabeth). parent(edward, albert). Etc.

Using ONLY the facts defined above, define rules in First Order Logic (FOL) using the Prolog for the following relations:

|                   |  |
|-------------------|--|
| mother(X, Y)      |  |
| daughter(X, Y)    |  |
| sibling(X, Y)     |  |
| grandMother(X, Y) |  |

- 15) [8 marks] Validate the following argument using propositional logic. Clearly state whether the argument is valid or invalid. Also, mention the rule of inference used.

- a. ARGUMENT#1:  
 Premise 1: If I study, I will pass.  
 Premise 2: I did not pass.  
 Conclusion: Therefore, I did not study.

|                   |  |
|-------------------|--|
| Inferencing rule? |  |
| Valid / invalid?  |  |

- b. ARGUMENT#2:  
 Premise 1: If it is winter, then it is cold.  
 Premise 2: It is cold.  
 Conclusion: Therefore, it is winter.

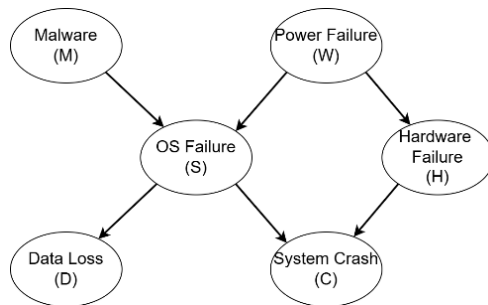
|                   |  |
|-------------------|--|
| Inferencing rule? |  |
| Valid / invalid?  |  |

### KNOWLEDGE REPRESENTATION (PROBABILISTIC REASONING)

16) [4 marks] For the Bayesian Network (BN) given here, assume each variable is binary.

Write the joint probability equation based on the structure of the graph.

Answer:



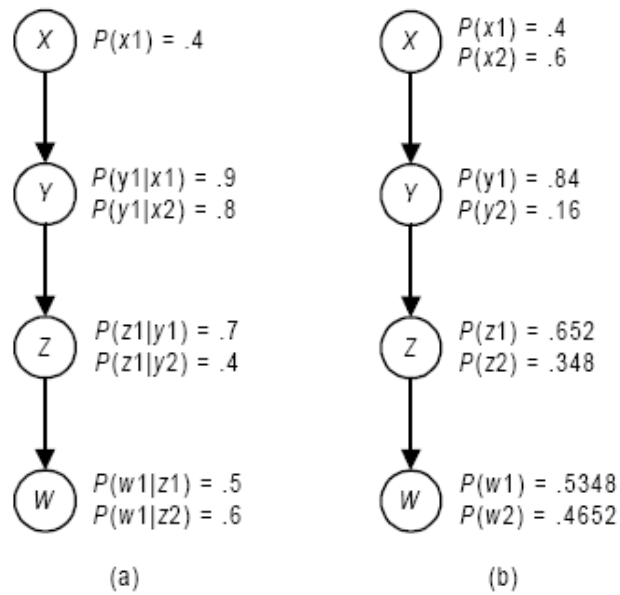
| M | W | P(S   M,W) |
|---|---|------------|
| T | T | 0.3        |
| T | F | 0.6        |
| F | T | 0.7        |
| F | F | 0.1        |

| W | P(H   W) |
|---|----------|
| T | 0.2      |
| F | 0.5      |

In the two tables above, you are given the conditional probabilities of for two variables (S and H). How many more (independent) probability values do we need to specify the joint probability distribution of this BN? Hint: think about the remaining variables.

Answer:

17) [8 marks] Suppose we have a singly connected Bayesian Network as shown in Figure (a) below. There are four variables: X, Y, Z, and W. The conditional probabilities are given in Figure (a), and the marginal probabilities of the four variables are given in Figure (b). Now, suppose we get evidence that z1 is true, i.e.,  $P(z1) = 1$ . Find the posterior probabilities of y1 and x1 (i.e.,  $P^*(y1)$  and  $P^*(x1)$ ).



Answers:

|                                              |  |
|----------------------------------------------|--|
| Posterior probability of y1 or $P^*(y1)$ is: |  |
| Posterior probability of x1 or $P^*(x1)$ is: |  |

18) [2 marks] A Bayesian Network factorizes the joint probability distribution into conditional probabilities of each variable given its parents. How do you return to the explicit joint distribution from this factorization? Circle the right option.

- a) Add the conditional probabilities of all variables given their parents.
- b) Multiply the conditional probabilities of each variable given its parents.
- c) Sum the probabilities of all nodes ignoring their parents.
- d) Multiply only the marginal probabilities of each variable.